LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Ch 10 Test (Set B): The Human Eye and the Colourful World

Chapter / Topic: The Human Eye and the Colourful World

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SECTION A

Q1. Which part of the eye controls the size of the pupil? **[1]**

Answer: (c) Iris

Q2. The splitting of white light into its component colours is called: **[1]**

Answer: (d) dispersion

Q3. The clear sky looks blue because of: **[1]**

Answer: (b) scattering of light

Q4. A person cannot see objects beyond 2 m clearly. The defect is: **[1]**

Answer: (a) myopia

Q5. Signals from the retina are carried to the brain by the: **[1]**

Answer: (b) optic nerve

Q6. **Assertion (A):** The sky appears dark to passengers flying at very high altitudes. **Reason (R):** At such heights there is very little scattering of… **[1]**

Answer: (a)

SECTION B

Q7. What are the near point and the far point of a normal human eye? **[2]**

Answer: Near point: the closest point at which the eye can see an object clearly without strain, about 25 cm for a young adult. Far point: the farthest point it can see clearly, infinity for a normal eye.

- Near point 25 cm: 1
- Far point infinity: 1

OR, answer: The image distance stays the same, because the image always forms on the retina. The eye adjusts by changing the focal length of the eye lens (accommodation), using the ciliary muscles.

- Image distance unchanged: 1
- Accommodation: 1

Q8. Why does the Sun appear white at noon but reddish at sunrise? **[2]**

Answer: At noon, sunlight travels a shorter distance through the atmosphere, so only a little of the blue and violet light is scattered and the Sun looks white. At sunrise, the light travels a much longer path near the horizon, so most blue light is scattered away and mainly red light reaches us.

- Noon: shorter path, less scattering: 1
- Sunrise: longer path, red reaches: 1

SECTION C

Q9. A person needs a lens of power −5.5 D to correct distant vision and a lens of power +1.5 D to correct near vision. **[3]**

(i) $f = 1/P = 1/(-5.5)$ m ≈ -0.18 m $= -18.2$ cm (concave lens) [1]

(ii) $f = 1/P = 1/1.5$ m $\approx +0.67$ m $= +66.7$ cm (convex lens) [1]

(iii) A bifocal lens: the upper part concave (for distant vision) and the lower part convex (for near vision). [1]

Q10. Why do different colours bend through different angles in a prism? What is a spectrum? Write the order of its colours. **[3]**

Answer: Different colours travel at slightly different speeds in glass, so they are refracted (bent) by different amounts. Red bends the least and violet the most. The band of coloured components of a light beam is called a spectrum. Order: Violet, Indigo, Blue, Green, Yellow, Orange, Red (VIBGYOR).

- Reason: 1
- Definition: 1
- Order: 1

OR, answer: Stars are very far away and act as point sources. Their light is refracted by layers of air whose refractive index keeps changing, so the star's apparent position and brightness flicker. Planets are much closer and act as extended sources, a collection of many point sources. The variations from all these points cancel out on average, so planets do not twinkle.

- Atmospheric refraction: 1
- Stars are point sources: 1
- Planets are extended sources, variations average out: 1

SECTION D

Q11. Answer the following: **[5]**

(i) Cornea: transparent front surface where most of the refraction of light happens. Iris: controls the size of the pupil. Pupil: regulates the amount of light entering the eye. Eye lens: fine-tunes the focus to form the image on the retina. Ciliary muscles: change the focal length of the lens (accommodation). Retina: light-sensitive screen on which the image forms. [3]

- Any six functions: ½ each

(ii) (Any two) Two eyes give a wider field of view: about 180° compared with about 150° for one eye. They improve our ability to detect faint objects. Each eye sees a slightly different image, and the brain combines them to give a sense of depth (three-dimensional vision). [2]

- Any two reasons: 1 + 1

OR, answer: (a) With age, the power of accommodation decreases because the ciliary muscles weaken and the eye lens becomes less flexible, so the near point moves away. A person may have both near and distant vision problems. A bifocal lens has a concave upper part for distant vision and a convex lower part for reading. (b) In old age the eye lens can become milky and cloudy, causing partial or complete loss of vision. Vision can be restored by cataract surgery.

- (a) cause 1, effect 1, bifocal 1
- (b) cloudy lens 1, surgery 1

SECTION E

Q12. Eyes can live even after death! By donating our eyes after we die, we can light up the life of a blind person. The eyes must be removed within 4–6… **[4]**

(i) Within 4–6 hours. [1]

(ii) The cornea. [1]

(iii) Up to four corneal-blind people. Reason: it costs the donor nothing after death but can give sight to people whose blindness is curable. [2]

- Up to four: 1
- Sensible reason: 1